Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech | *Subject: Structural Analysis* | *Time: 03 Hours*
Code: 101408 | *Full Marks: 70*

Instructions:-

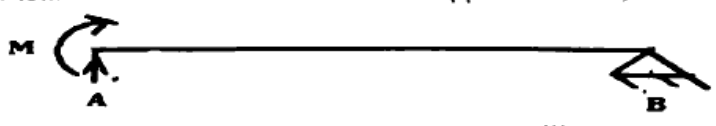
(i) The marks are indicated in the right-hand margin
*(ii) There are **NINE** questions in this paper*
*(iii) Attempt **FIVE** questions in all*
(iv) Question No. 1 is compulsory

Q.1 Choose the correct option / write the answer of the following (Any seven): **[2 x 7 = 14]**

(a) Which of the following is carried by truss members?
(i) flexural load (iii) axial load
(ii) bending load (iv) shear load

(b) When a structural member of the uniform section is subjected to a moment at one end only, then the moment required so as to rotate that end to produce a unit slope, is called
(i) Stiffness of member (iii) Potential of member
(ii) Capacity of member (iv) Resistance of member

(c) Carryover Moment at end B due to moment M applied at end A for the given beam is



(i) -M (iii) +M
(ii) +M2 (iv) 0

(d) Which of the following methods of structural analysis is a force method?
(i) three moment equation
(ii) slope deflection method
(iii) column analogy method
(iv) moment distribution method

(e) Force-displacement requirement of statically indeterminate structures depend upon which factor?
(i) supports (iii) position of load applied
(ii) material's response (iv) magnitude of load applied

(f) In influence line diagrams (ILD) _____
(i) Points remain fixed, position of load changes
(ii) Points change, position of loads remain fixed
(iii) Both of them changes
(iv) Neither of them changes

(g) Maxwell's reciprocal theorem is applicable for only prismatic members.
(i) Prismatic (iii) Both
(ii) Non prismatic (iv) None of these

(h) If a structure has total 10 joints, then what should be the minimum no. of joints in which equilibrium equations should be concurrently satisfied for stability?
(i) 9 (iii) 8
(ii) 10 (iv) 7

(i) Betti's Theorem is based on _____
(i) Balancing of external and internal forces
(ii) Balancing of work done produced by external and internal loadings
(iii) Balancing of external and internal moments
(iv) Balancing of strain energy produced by external and internal loadings

(j) Flexibility matrix is always:-
(i) symmetric (iii) anti-symmetric
(ii) non-symmetric (iv) depends upon loads applied

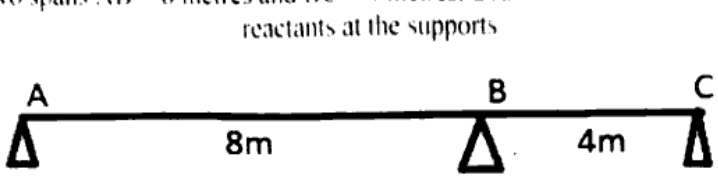
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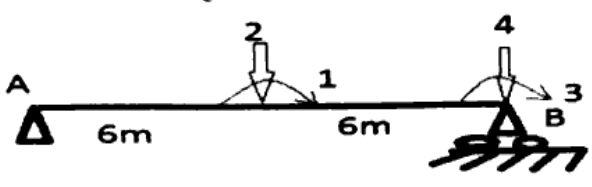
Q.2 (a) Write the Comparison between Force Method and Displacement method. **[7]**

(b) A two-hinged parabolic arch hinged at the springs at the same level, has a horizontal span of 20 m and a central rise of 2 m. Draw influence line diagrams for (i) Horizontal thrust at support (ii) B.M. at a section 5 m from the left support. Hence or otherwise find the maximum positive and negative bending moments at the above section due to the passage of a point load of 100 kN over the span. Assume that the moment of inertia of the arch section varies as the secant of the slope. **[7]**

Q.3 (a) A continuous beam ABC is hinged at A and C and simply supported at B. It consists of two spans AB = 8 metres and BC = 4 metres. Draw influence lines for the reactants at the supports **[7]**



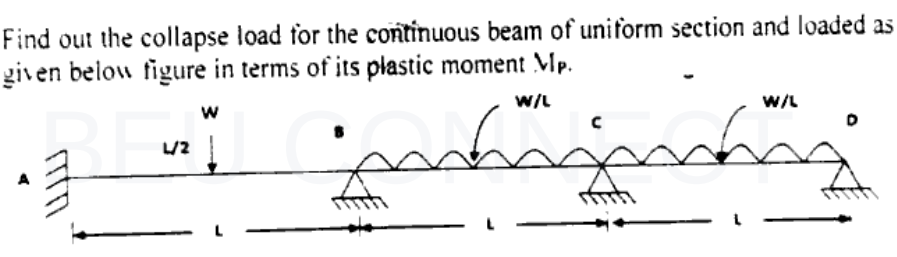
(b) Develop the flexibility matrix for the simply supported beam AB with reference to the coordinates shown in fig. **[7]**



Q.4 (a) State Muller Breslau's principle. **[7]**

(b) What is meant by shape factor and also mention the shape factor for (i) circular and (ii) triangular sections. **[7]**

Q.5 (a) Find out the collapse load for the continuous beam of uniform section and loaded as given below figure in terms of its plastic moment M_P. **[9]**

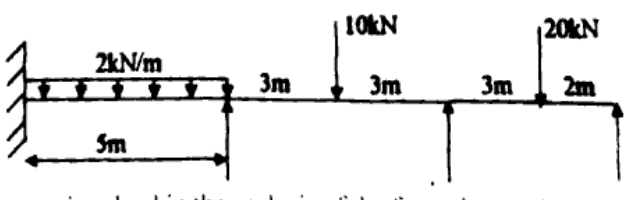


(b) Discuss the detailed design steps for the design of column with axial load and bi axial bending. **[5]**

Q.6 (a) A continuous beam ABC consists of spans AB and BC of 5 m length in each. Both the ends of the beam are fixed. The span AB carries a point load of 15 kN at its middle point. The span BC carries a point load of 25 kN at its middle point. Find the moments and reactions at the supports. Assume the beam is of uniform section. Use Slope deflection method. **[7]**

(b) State the difference between strain energy method and unit load method in the determination of deflection of structures. **[7]**

Q.7 (a) Draw the bending moment diagram and shear force diagram for the continuous beam shown in figure below using moment distribution method. EI is constant **[7]**



(b) Explain the steps involved in the analysis of the frame by cantilever method **[7]**

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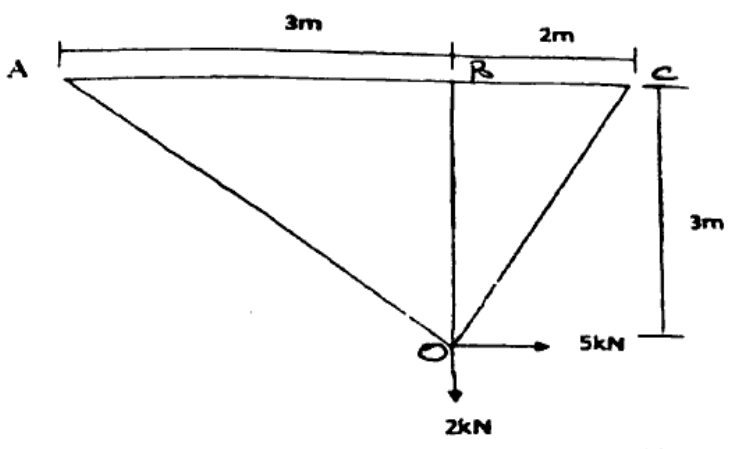
Q.8 (a) Explain the principle of virtual work and its significance in structural analysis. **[7]**

(b) Compare the behaviour of a three hinged arch and a fixed arch under loading conditions. **[7]**

Q.9 Answer ***any two*** of the following:- **[7x2=14]**

(a) Explain the steps involved in the determination of deflection of pin jointed plane frames and rigid plane frames.

(b) Analyse the truss show in figure by stiffness method.



(c) Explain the castigliano'second theorem and apply it to simply supported beam under a concentrated load.

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